



User Manual

PathHunter® CHO-K1 NPY2R Bioassay Kit

For the Measurement of Ligand-Mediated Arrestin Recruitment

For Bioassay Kits:

93-0212Y2-00065: 2-Plate Kit

93-0212Y2-00066: 10-Plate Kit



Document Number 70-455 Revision 1

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1. Overview

The PathHunter CHO-K1 NPY2R Bioassay kit provides a robust and highly sensitive functional, cell-based assay to monitor β -arrestin-2 recruitment as a result of ligand-mediated NPY2R activation. The bioassay kit contains all the materials needed for a complete assay, including cryopreserved, ready-to-use cells, detection reagents, cell plating reagent, dilution buffer, positive control agonist, and assay plates. The ready-to-use cryopreserved cells have been manufactured to ensure assay reproducibility and faster implementation from characterization to lot release. This bioassay has been optimized for a 96-well format but can be adapted to a 384-well format if needed.

2. Assay Principle

The PathHunter CHO-K1 NPY2R Bioassay monitors NPY2R activity by detecting the interaction of β -Arrestin-2 with the activated GPCR using β -galactosidase (β -gal) enzyme fragment complementation. In this system, the NPY2R is fused, in frame, with the small β -gal fragment called ProLink™ (PK) and co-expressed in cells stably expressing a fusion protein of β -Arrestin-2 with the larger β -gal fragment, enzyme acceptor (EA). Activation of NPY2R stimulates the recruitment of β -Arrestin-2 to the ProLink-tagged receptor and forces complementation of the two enzyme fragments of β -galactosidase. This results in the formation of an active β -gal enzyme leading to an increase in enzyme activity that can be measured using the chemiluminescent PathHunter Bioassay Detection Reagents. Because arrestin recruitment occurs independent of G-protein coupling, these assays provide a direct, universal platform for measuring receptor activation, thus providing an ideal tool to characterize functional biologics.

β -Arrestin recruitment

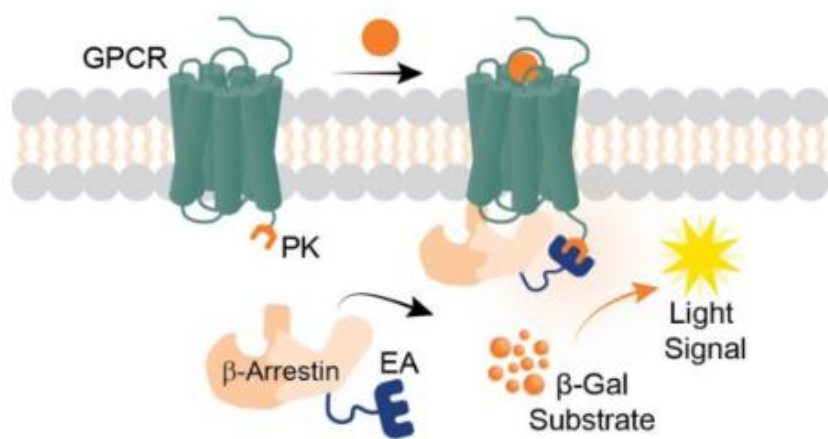


Figure 1. Assay Principle for the PathHunter CHO-K1 NPY2R Bioassay.

3. Materials Provided

List of Components	93-0212Y2-00065 (2-plate Kit)	93-0212Y2-00066 (10-plate Kit)
PathHunter CHO-K1 NPY2R Bioassay Cells (1.2×10^6 cells in 0.1 mL per vial)*	2 Vials	10 Vials
AssayComplete™ Cell Plating 0 Reagent (CP0) (100 mL per bottle)	1 x 100 mL	4 x 100 mL
AssayComplete Protein Dilution Buffer (50 mL per bottle)	1 x 50 mL	2 x 50 mL
Control Agonist (Peptide YY, Human)** (500 µg per vial)	1 x 500 µg	1 x 500 µg
PathHunter Bioassay Detection Kit		
Detection Reagent 1 (Bottle)	1 x 3 mL	1 x 15 mL
Detection Reagent 2 (Bottle)	1 x 12 mL	1 x 60 mL
96-well White, Clear bottom, TC-treated Sterile Plates with Lid	2 Plates	10 Plates

***Note:** More cells are provided per vial than are needed in the assay. Excess cells may be discarded after plating.

****Note:** The positive control is used for QC testing of the bioassay cells in this kit, as reflected in the Certificate of Analysis

4. Storage Conditions

PathHunter CHO-K1 NPY2R Bioassay Cells

Cells are shipped on dry ice and should arrive in a frozen state. To ensure maximum cell viability, store the vials of bioassay cells in the vapor phase of liquid nitrogen as soon as possible upon receipt. Please contact technical support immediately, if the cells received were already thawed.

- Short-term (24 hours or less): Store vials at -80°C immediately upon arrival. (DO NOT store at -80°C for longer than 24 hours).
- Long-term (greater than 24 hours): Vials should ONLY be stored in the vapor phase of liquid nitrogen.



Safety Warning: A face shield, gloves and lab coat should be worn at all times when handling frozen vials. Use tongs to remove cryovials from liquid nitrogen storage and place the vials immediately on dry ice in a covered container. Wait at least 1 minute for any liquid nitrogen that may be present inside the vial to evaporate. Do not touch the bottom of the vials at any time to avoid inadvertent thawing of the cells.

AssayComplete™ Cell Plating 0 Reagent (CP0)

Upon receipt, store at -20°C. Once thawed, the Cell Plating 0 Reagent can be stored at 4°C for up to 4 weeks. For longer storage (up to the expiration date listed on the kit's Certificate of Analysis), the reagent

should be aliquoted and stored at -20°C. Once aliquoted, do not freeze-thaw more than two times. Make aliquots suitable for testing one assay plate, for example: 40 mL of reagent per aliquot can be dispensed and stored at -20°C. Ensure that the reagent is equilibrated to room temperature before use in the assay for best performance.

AssayComplete Protein Dilution Buffer (PDB)

Upon receipt, store at -20°C. Once thawed, the Protein Dilution Buffer can be stored at 2-8°C. for up to 4 weeks. For longer storage (up to the expiration date listed in the kit's Certificate of Analysis), the reagent should be aliquoted and stored at -20°C until needed. Avoid multiple freeze-thaw cycles. To make aliquots suitable for testing one assay plate each, 10 mL of reagent per aliquot can be dispensed and frozen down. This may vary depending on your stock sample concentrations and should be adjusted accordingly. Ensure that the reagent is equilibrated to room temperature before use in the assay for best performance.

Control Agonist (Peptide YY, Human)

Upon receipt, store at -20°C until ready to use (up to the expiration date listed on the kit's Certificate of Analysis). To avoid condensation, equilibrate the vial to ambient temperature before opening. Centrifuge the vial prior to opening, to maximize recovery, and reconstitute by adding 580 µL sterile water to the 500 µg in the vial to generate an 862 µg/mL (200 µM) stock¹⁵. Once prepared, the stock solution should be stored as suitable aliquots (e.g., 30 µL) at -20°C until needed (avoid multiple freeze thaw cycles). The reconstituted ligand is stable for 12 months at -20 to -80°C, or for 1 week at 2-8°C.

PathHunter Bioassay Detection Kit

Upon receipt, store reagents at -20°C for long-term storage, up to the indicated shelf life of that component. Thaw reagents at room temperature before use. Thawed reagents are stable for up to 4 days when stored at 2-8°C. For long-term storage (up to the expiration date listed in the kit's Certificate of Analysis), the reagent should be re-frozen in opaque containers at -20°C until needed. Avoid multiple freeze-thaw cycles.

For the ten-plate kit, if all the plates will not be used at the same time, we recommend making five aliquots for each of the two detection reagents. Each aliquot will be adequate for two 96-well assay plates. Make five aliquots of 2.3 mL each for Detection Reagent 1, and five aliquots of 9.2 mL each for Detection Reagent 2. Sufficient reagent volumes are provided in the kit to make these aliquots. If reagents will be used for a single plate, then the remaining Detection Reagents can be frozen. The detection reagents can be thawed and frozen for a total of three times without loss in performance.

96-Well Clear Bottom Tissue Culture-Treated Plates

Upon receipt, store at room temperature.

5. Additional Equipment and Materials Required

The following equipment and additional materials are required to perform these assays:

Material	Ordering Information
96-Well Green, V-Bottom, Untreated, Non-Sterile Dilution Plates	DiscoverX, Cat No. 92-0011, or similar
Multimode or luminescence plate reader	Refer to the Instrument Compatibility Chart at discoverx.com/instrument-compatibility
Sterile disposable reagent reservoir	Thermo Fisher Scientific, Cat. No. 8094, or similar
50 mL and 15 mL Polypropylene tubes, sterile	
1.5 mL polypropylene microcentrifuge tubes, sterile	
Tissue culture disposable pipettes (1 mL - 25 mL), sterile	
Single and multichannel micropipettes	
Disposable pipet tips for P20, P100, P1000 micropipettes (10 µL-1000 µL)	
Sterile biosafety cabinet	
Humidified tissue culture incubator (37°C, 5% CO ₂)	
Water bath (set to 37°C)	

6. Unpacking Cell Cryovials

Cryovials are shipped on dry ice and contain cells in 0.1 mL of AssayComplete™ Freezing Reagent (refer to the CoA for cell number provided in the vial). Upon receipt, the vials should be transferred to the vapor phase of liquid nitrogen for storage beyond 24 hours. The following procedures are for the safe storage and removal of cryovials from liquid nitrogen storage.



Appropriate personal protective equipment (PPE), including a face shield, gloves, and a lab coat should be worn during these procedures.

1. PathHunter NPY2R Bioassay Cells must arrive in a frozen state on dry ice.



Contact technical support immediately, if cells received were already thawed.

2. Frozen cells must be transferred to either liquid nitrogen storage or a -80°C freezer immediately upon arrival. If cells will be thawed and used within 24 hours, they can be stored temporarily at -80°C. For long-term storage, place vials in the vapor phase of liquid nitrogen storage.



Cryovials should be stored in the vapor phase of liquid nitrogen ONLY. These vials are not rated for storage in the liquid phase.

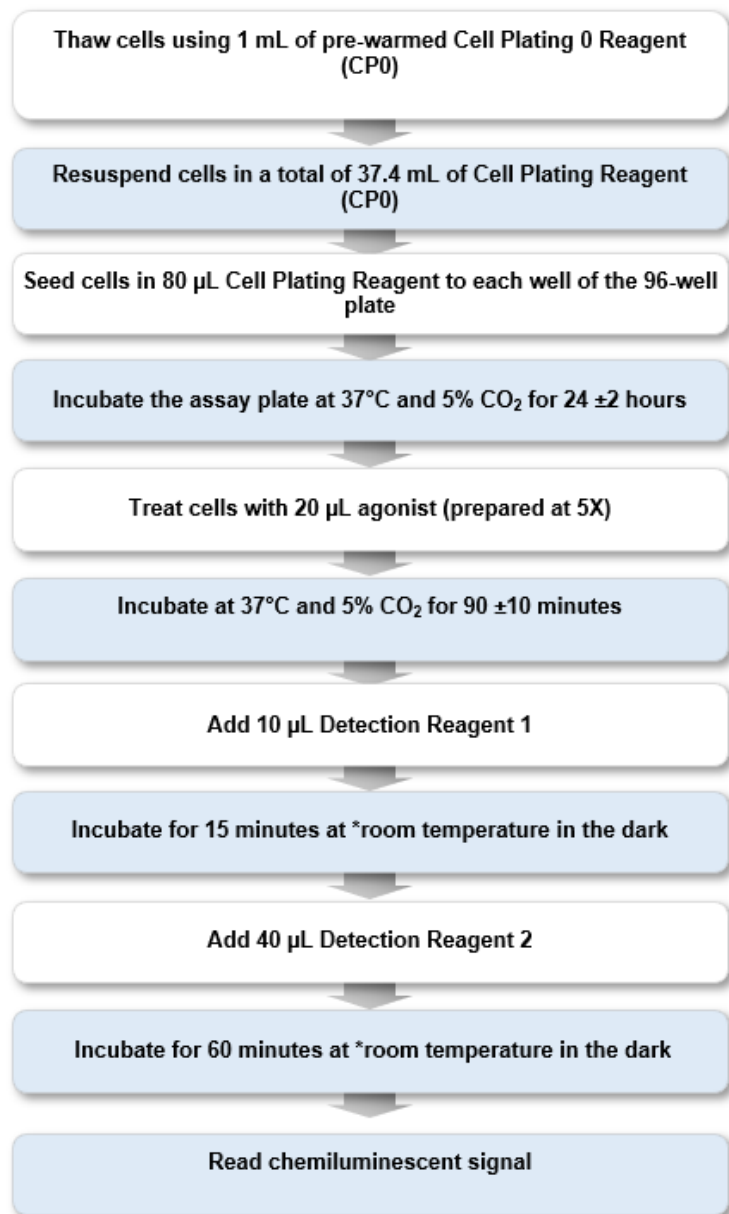
3. Using tongs, remove the cryovials from the liquid nitrogen storage, and place them Immediately in

dry ice, in a covered container. Wait for at least one minute for any liquid nitrogen inside the vial to evaporate.

4. Proceed with the thawing protocol in the [Bioassay Cell Preparation](#) section of the [Detailed Assay Protocol](#) section.

7. Protocol Schematic

Quick-Start Procedure: In a 96-well tissue culture treated plate perform the following steps.



*Room temperature refers to a range of 23-25°C

8. Detailed Protocol

This user manual provides a protocol for testing samples in the PathHunter CHO-K1 NPY2R bioassay in a 96-well format. Prepared volumes may be scaled up or down if required.

This assay is performed under aseptic conditions. Cells and reference standards/test samples are prepared in a Biological Safety Cabinet following good aseptic techniques. All appropriate materials are either certified sterile or prepared aseptically.

8.1 Bioassay Cell Preparation

Day 1:

The following protocol is for thawing and plating cryopreserved CHO-K1 NPY2R bioassay cells from cryovials.

1. Before thawing the cells, ensure that all the materials required are set up in the biosafety cabinet. These include:
 - a. One sterile 25 mL reagent reservoir
 - b. One sterile 50 mL conical tube
 - c. A micropipette (P1000) set to dispense 1 mL
 - d. A multichannel pipette and tips set to dispense 80 µL
 - e. A bottle of AssayComplete™ Cell Plating 0 Reagent (CP0), **pre-warmed in a 37°C water bath for 15 minutes and equilibrated to room temperature.**
 - f. A 96-Well White, Clear-bottom TC-treated Sterile Assay Plate (provided with the kit)
2. Dispense 38.4 mL of CP0 into the 50 mL conical tube.
3. Remove the cryovial from the liquid nitrogen vapor and immediately place it in dry ice.
4. Remove the cryovial from dry ice and ensure cap is tightened. Immediately thaw vial in 37°C water bath for 30 (+/- 5) seconds, gently shaking to thaw cells. **DO NOT LEAVE IN WATER BATH.**
5. Visually inspect bottom of vial after 20 seconds. If pellet is thawed, remove vial from water bath, wipe down the outside surface of the cryovial quickly with 70% ethanol, and immediately bring it into the biosafety cabinet. If ice is still visible, return vial to water bath for additional 10-15 seconds. Add 1 mL of pre-warmed CP0 from the 50 mL conical tube into the cryovial. The reagent should be added slowly along the wall of the cryovial tube. Slowly pipet the cell suspension up and down, several times, to uniformly resuspend the cells.
6. Transfer the cell suspension to the conical tube containing the remaining 37.4 mL of CP0 (resulting in a cell density of 31,250 cells/mL). Remove any remaining liquid from the cryovial to ensure maximum recovery of all the cells.
7. Replace the cap on the conical tube and mix by gentle inversion 2-3 times to ensure that the cells are properly resuspended in the reagent, without creating any froth in the suspension. Immediately transfer the suspension into the sterile 25 mL reagent reservoir.
8. Using a manual 12-channel multichannel pipette, transfer 80 µL of the cell suspension (2,500 cells / well) to each well of the 96-well assay plate one row at a time, using reverse pipetting. Mix cells

in reagent reservoir by pipetting up/down 2-3 times before aspirating and pipetting cells into each subsequent row in the assay plate.

Note: Any remaining cell suspension may be discarded.

9. Replace the lid and leave the plate at room temperature for 15 minutes, to allow the cells to settle uniformly in the well and minimize potential edge effects.
10. Incubate the assay plate in humidified tissue culture incubator at 37°C and 5% CO₂ for 24 ± 2 hours before proceeding with the assay.

8.2 Agonist Sample Preparation (Peptide YY and Test Samples)

Day 2:

The following protocol gives the recommended dilutions for preparing the Peptide YY control agonist and test sample working stocks and serial dilutions.

1. Before beginning, ensure all required reagents are assembled in biosafety cabinet:
 - a. Micropipettes (P20, P100, P1000)
 - b. A multichannel pipette and tips set to dispense 20 µL.
 - c. An aliquot of AssayComplete™ Protein Dilution Buffer (PDB), **pre-warmed in a 37°C water bath for 15 minutes, then equilibrated to room temperature.**
 - d. A 96-Well Green, V-Bottom, Untreated, Non-Sterile Dilution Plate, labeled 'Master dilution plate' (MDP)
 - e. Sterile microfuge tubes, for preparation of intermediate, working stocks of Peptide YY.
2. On the day of the assay, prepare Peptide YY intermediate and working stock solutions in PDB as shown in examples in Table 1 below.
 - a. Peptide YY, human is supplied as a lyophilized powder. To avoid condensation, equilibrate the vial to ambient temperature before opening. Centrifuge the vial prior to opening, to maximize recovery. Reconstitute by adding 580 µL of sterile water to the 500 µg in the vial to generate an 862 µg/mL (200 µM) stock.
 - i. Reconstituted Peptide YY stock from 2a may be stored in small aliquots at -20°C.
 - b. It is recommended to use a minimum aliquot size of 10 µL for preparation of all sample / working stock dilutions (as shown in Table 1).

Table 1. Example Preparation of Peptide YY Intermediate & Working Stocks

Intermediate and Working Stock Solutions	Final Concentration, µg/mL	Volume of Peptide YY (Stock concentration)	Volume Dilution Buffer (PDB), µL
Peptide YY, Intermediate Stock	65	10 µL (of 862 µg/mL stock)	123
Peptide YY Working Stock	6.5	20 µL (of 65 µg/mL stock)	180

Note: Use a vortex to mix (10-20 sec) for each dilution before preparing next serial dilution.

3. On the day of the assay, prepare serial dilutions of the Peptide YY in row A of the 96-well master dilution plate (MDP), at 5X the final concentration of each dilution, in PDB (i.e., the Ligand Diluent) as per Table 2 below. An example sample dilution scheme is indicated in Table 2 below. Following this dilution scheme should result in sufficient volumes to run triplicate wells per dose in the assay plate.
- Starting at Row A, Well 1 of the MDP, add appropriate volume of ligand diluent (PDB) as indicated in column 6 of Table 2.
 - Transfer appropriate volume (indicated in column 5 of Table 2) of Peptide YY Working Stock to the MDP **Row A, Well 2**.
 - Prepare the dilution series by transferring the indicated volume of Peptide YY sample (column 5, Table 2) from the source well (indicated in column 4) to the destination well (indicated in column 1). Pipette up and down several times to mix in destination wells. Replace pipet tips between each serial dilution.
 - No Peptide YY working stock is added to well 1 (vehicle only), as this serves as the negative control.

Table 2. Sample concentration in Master Dilution Plate prepared as a 5X stock: Example of Preparation of Peptide YY Control Serial Dilutions

Column 1	Column 2	Column 3	Column 4	Column 5	Column 6	Column 7
Well location on Master Dilution Plate (MDP)	Concentration (5X) of Peptide YY on MDP Row A, ng/mL	Dilution Factor	Dilution (5X) Peptide YY Source Well	Volume (5X) of Peptide YY added, μ L	Volume of Ligand Diluent, μ L	Final Concentration (1X) of Peptide YY in Assay Plate, ng/mL
Row A, Well 1	0	--	--	--	180	--
Row A, Well 2	6,500	--	Working Stock (6.5 μ g/mL; see Table 1)	200	-	1,300
Row A, Well 3	2166.7	3	Row A, Well 2	100	200	433.3
Row A, Well 4	722.2	3	Row A, Well 3	100	200	144.4
Row A, Well 5	240.7	3	Row A, Well 4	100	200	48.1
Row A Well 6	80.2	3	Row A, Well 5	100	200	16.0
Row A, Well 7	26.75	3	Row A, Well 6	100	200	5.35
Row A, Well 8	8.92	3	Row A, Well 7	100	200	1.78
Row A, Well 9	2.97	3	Row A, Well 8	100	200	0.59
Row A, Well 10	0.991	3	Row A, Well 9	100	200	0.198
Row A, Well 11	0.330	3	Row A, Well 10	100	200	0.066
Row A, Well 12	0.110	3	Row A, Well 11	100	200	0.022

4. On the day of the assay, if Test Sample (TS) is an agonist, prepare appropriate dilutions of TS stock as needed, to generate 200 µL of a working stock of appropriate concentration, that will represent 5X the top desired concentration to be tested in the dose response curve for the TS.

Note 1: Appropriate concentration range and dilution factors for client TS should be determined empirically.

Note 2: Depending on stock concentration of TS, intermediate dilutions of TS may need to be prepared (as demonstrated in Table 1 for the Peptide YY control) to generate the TS working stock, which will be used as the top dose in the dose response curve in Step 5 below.

- a. All dilutions should be prepared in PDB diluent. Volumes may be scaled up or down as needed. However, it is recommended to use a minimum aliquot size of 10 µL for preparation of all sample / stock dilutions.
5. On the day of the assay, prepare serial dilutions of TS in row B of the 96-well master dilution plate (MDP), at 5X the final concentration of each dilution, in PDB (i.e., the Ligand Diluent). Volumes similar to those used in the example in Table 2 will be sufficient to run triplicate wells per dose in the assay plate.
 - a. Note: start top dose of DRC in Column 2, as for Peptide YY example in Table 2.
6. Transfer 20 µL of the 5X sample dilutions from the MDP to the appropriate wells containing cells in the assay plate. Refer to example in Figure 2. Representative Assay Plate Map:
 - a. Row A in MDP: transfer to Rows B, D and F in the assay plate (Peptide YY control)
 - b. Row B in MDP: transfer to Rows C, E and G in the assay plate (Test Sample)
7. Replace lid on assay plate and gently mix the contents by slowly moving the plate back and forth 3-4 times on the surface of the biosafety cabinet, in a criss-cross pattern.
8. Incubate the assay plate in a humidified incubator at 37°C and 5% CO₂ for 90 ± 10 minutes.
9. Proceed to Signal Detection section.

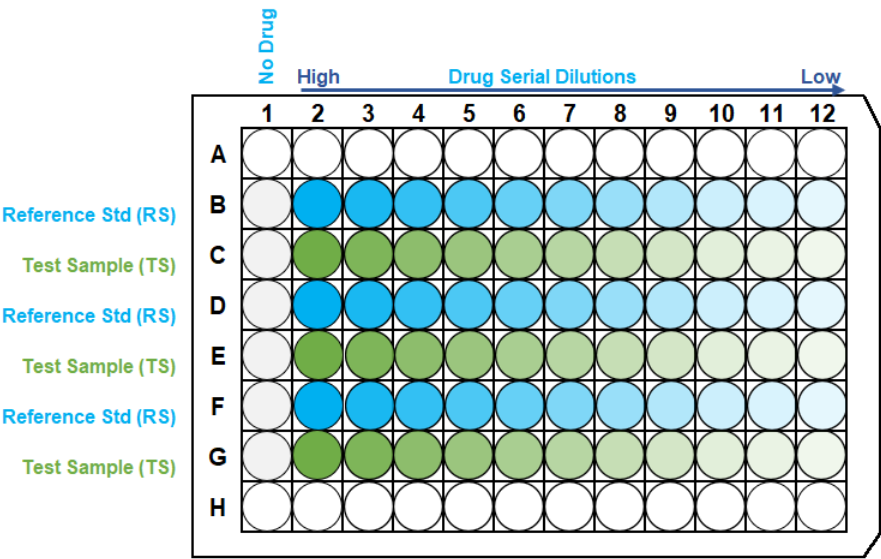


Figure 2. Representative Assay Plate Map

This plate map shows two interdigitated 11-point dose curves, with three replicates per dose point, for a test sample (TS) and control agonist (shown as reference standard in figure). Rows A and H contain cells only but may be used for an assay control with appropriate validation. Column 2 contains the highest dose of each sample, while column 12 contains the lowest dose. Column 1 contains no drug (ligand diluent only).

8.3 Addition of Detection Reagent

Signal Detection: Day 2

1. Thaw one aliquot each of Bioassay Detection Reagent 1 and Bioassay Detection Reagent 2 from PathHunter Bioassay Detection kit; equilibrate to room temperature.
2. Transfer 2.3 mL of Bioassay Detection Reagent 1 into a sterile 25 mL reservoir.
3. Remove assay plate from incubator and remove lid. Using a multichannel pipette, add 10 µL of the Bioassay Detection Reagent 1 from the reservoir into each well of the assay plate.
4. Replace lid on assay plate and gently mix the contents by slowly moving the plate back and forth 3-4 times on the surface of the biosafety cabinet, in a criss-cross pattern. **Do not pipette up and down or vortex the plate to mix.**
5. Incubate the assay plate for 15 minutes (+/- 5 minutes) at room temperature in the dark (e.g., in a drawer or covered location on the bench top).

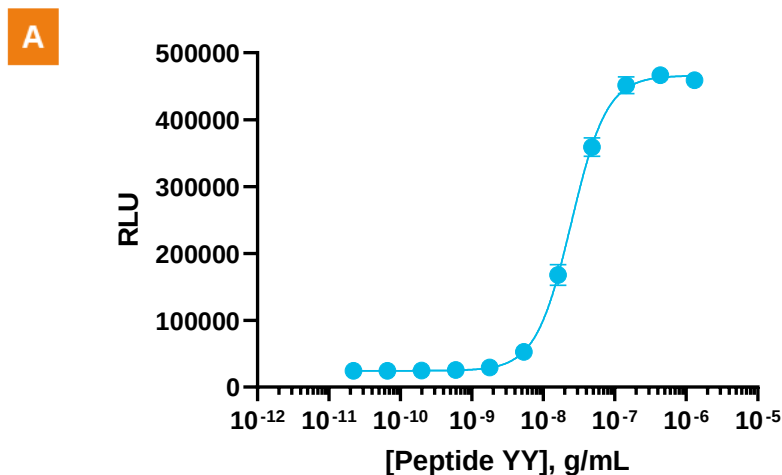


Room temperature refers to a range of 23-25°C.

6. Transfer 9.2 mL of Bioassay Detection Reagent 2 into a fresh sterile reservoir.
7. Remove assay plate from incubator and remove lid. Using a multichannel pipette, add 40 µL of the Bioassay Detection Reagent 2 from the reservoir into each well of the assay plate.
8. Replace lid on assay plate and gently mix the contents by slowly moving the plate back and forth 3-4 times on the surface of the biosafety cabinet, in a criss-cross pattern. **Do not pipette up and down or vortex the plate to mix.**
9. Incubate the assay plate for 1 hour (+/- 15 minutes) at room temperature in the dark (e.g. in a drawer or covered location on the bench top).
10. Read assay plate on a Perkin Elmer Envision, SpectraMax iD3, or a similar plate reader, in luminescence mode with 0.2 second/well integration time. Refer to the Instrument Compatibility Chart at discoverx.com/instrument-compatibility/
11. Data analysis can be performed using any statistical analysis software such as GraphPad Prism, SoftMax Pro, Gen5, or Microsoft Excel etc, or other preferred software.

9. Typical Results

The following graph is an example of a typical dose-response curve for the PathHunter CHO-K1 NPY2R Bioassay generated using the protocol outlined in this manual. The data shows a potent, dose-dependent increase in β -arrestin recruitment when treated with Peptide YY. The plate was read on a Molecular Devices Spectramax iD3 plate reader, with a 0.2 sec/well integration time, and data analysis was conducted using GraphPad Prism.



B

S/B	EC ₅₀ , ng/mL
18.9	24.6

Typical Results. Representative **A**, agonist dose-response curve and **B**, EC₅₀ and assay window (S/B) for Peptide YY mediated NPY2R activation, as measured in this bioassay.

10. Troubleshooting Guide

Problem	Potential Cause	Proposed Solution
No response	Incorrect thawing procedure	Refer to the thawing instructions in the Bioassay Cell Preparation section of this user manual.
	Incorrect ligand used or improper ligand incubation time	Refer to the Certificate of Analysis for recommended ligand and assay conditions.
	Incorrect preparation of the ligand (agonist or antagonist) Sub-optimal time course for induction	Refer to the vendor-specific datasheet to ensure proper handling, dilution, and storage of the ligand. Optimize time course of induction with agonist and antagonist.
Decreased or no response	Cells are not adherent and exhibit incorrect morphology	Confirm adherence of cells using a microscope.
	Problem with microplate reader	The microplate reader should be in luminescence mode. Read at 0.1-1 seconds/well.
Low or no signal	Incorrect preparation of detection reagents	Detection reagents are sensitive to light and should ideally be prepared just prior to use.
	Problem with microplate reader	The microplate reader should be in luminescence mode. Read at 0.1-1 second/well.
Experimental S/B does not match the value noted in the Certificate of Analysis provided	Incorrect incubation temperature	Confirm assay conditions. Check and repeat the assay at the correct incubation temperature, as indicated on the assay datasheet.
	Incorrect preparation of ligand (agonist or antagonist)	Some ligands are difficult to handle. Confirm the final concentration of ligands.
EC ₅₀ is right-shifted	Improper ligand handling or storage	Ensure that the ligands are stored and incubated at the proper temperature.
	Difference in agonist binding affinity	Confirm that the ligand used is comparable to the ligand in the Certificate of Analysis.
	Problems with dynamic range or dilutions	Changing tips during dilution can help in avoiding carryover. Include a cAMP standard curve in the assay to ensure that the ligand dose curve is in the dynamic range of the cAMP detection kit. Moving out of the dynamic range may shift the EC ₅₀ of the ligands.
High variability between replicates	Instrument calibration	Ensure that the dispensing equipment is correctly calibrated, and proper pipetting technique is used.

For questions on using this product, please contact Technical Support at discoverx.com/support/

Document Revision History

Revision Number	Date Released	Revision Details
0	June 2024	New Document
1	July 2025	Materials tables update

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